```python
import pandas as pd
df = pd.read_csv(r"C:\Users\uktiw\Downloads\Python\matches -
matches.csv")
df.head()
```

```
   id  season       city        date                          team1  \
0   1    2017  Hyderabad  2017-04-05           Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06                Mumbai Indians
2   3    2017     Rajkot  2017-04-07                 Gujarat Lions
3   4    2017     Indore  2017-04-08        Rising Pune Supergiant
4   5    2017  Bangalore  2017-04-08  Royal Challengers Bangalore

                          team2                  toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat

   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
3               6      GJ Maxwell                     Holkar Cricket
Stadium
4               0       KM Jadhav                      M Chinnaswamy
Stadium

          umpire1        umpire2  umpire3
0     AY Dandekar       NJ Llong      NaN
1  A Nand Kishore         S Ravi      NaN
2     Nitin Menon      CK Nandan      NaN
```

```
3    AK Chaudhary  C Shamshuddin      NaN
4             NaN            NaN      NaN
```

How many matches are there

```
df.shape[0]

636
```

How many attributes per match?

```
df.shape[1]

18
```

What kind of information is present?

```
df.columns

Index(['id', 'season', 'city', 'date', 'team1', 'team2',
'toss_winner',
       'toss_decision', 'result', 'dl_applied', 'winner',
'win_by_runs',
       'win_by_wickets', 'player_of_match', 'venue', 'umpire1',
'umpire2',
       'umpire3'],
      dtype='object')
```

Structure of data

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 636 entries, 0 to 635
Data columns (total 18 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   id               636 non-null    int64
 1   season           636 non-null    int64
 2   city             629 non-null    object
 3   date             636 non-null    object
 4   team1            636 non-null    object
 5   team2            636 non-null    object
 6   toss_winner      636 non-null    object
 7   toss_decision    636 non-null    object
 8   result           636 non-null    object
 9   dl_applied       636 non-null    int64
 10  winner           633 non-null    object
 11  win_by_runs      636 non-null    int64
```

```
 12  win_by_wickets   636 non-null    int64
 13  player_of_match  633 non-null    object
 14  venue            636 non-null    object
 15  umpire1          635 non-null    object
 16  umpire2          635 non-null    object
 17  umpire3          0 non-null      float64
dtypes: float64(1), int64(5), object(12)
memory usage: 89.6+ KB
```

Statistical information about data

```
df.describe()
```

```
                id        season  dl_applied  win_by_runs
win_by_wickets  \
count  636.000000   636.000000  636.000000   636.000000
636.000000
mean   318.500000  2012.490566    0.025157    13.682390
3.372642
std    183.741666     2.773026    0.156726    23.908877
3.420338
min      1.000000  2008.000000    0.000000     0.000000
0.000000
25%    159.750000  2010.000000    0.000000     0.000000
0.000000
50%    318.500000  2012.000000    0.000000     0.000000
4.000000
75%    477.250000  2015.000000    0.000000    20.000000
7.000000
max    636.000000  2017.000000    1.000000   146.000000
10.000000

       umpire3
count      0.0
mean       NaN
std        NaN
min        NaN
25%        NaN
50%        NaN
75%        NaN
max        NaN
```

Which columns have missing values

```
df.isnull().sum()[df.isnull().sum() >
0].sort_values(ascending=False).head(6)
```

```
umpire3            636
city                 7
```

```
winner                3
player_of_match       3
umpire1               1
umpire2               1
dtype: int64
```

```
df.isnull().sum()[df.isnull().sum() > 0]
```

```
city                  7
winner                3
player_of_match       3
umpire1               1
umpire2               1
umpire3             636
dtype: int64
```

Which column is most unreliable due to missing data

```
df.isnull().sum().sort_values(ascending=False).head(1)
```

```
umpire3    636
dtype: int64
```

What is the biggest wins by runs ever?

```
df["win_by_runs"].sort_values(ascending=False).head(1)
```

```
43    146
Name: win_by_runs, dtype: int64
```

Which team won the most matches?

```
df["winner"].value_counts().head(1)
```

```
winner
Mumbai Indians    92
Name: count, dtype: int64
```

How many cities hosted IPL?

```
df['city'].nunique()
```

```
30
```

Which cities hosted IPL?

```
df['city'].unique()
```

```
array(['Hyderabad', 'Pune', 'Rajkot', 'Indore', 'Bangalore', 'Mumbai',
       'Kolkata', 'Delhi', 'Chandigarh', 'Kanpur', 'Jaipur',
'Chennai',
       'Cape Town', 'Port Elizabeth', 'Durban', 'Centurion',
       'East London', 'Johannesburg', 'Kimberley', 'Bloemfontein',
       'Ahmedabad', 'Cuttack', 'Nagpur', 'Dharamsala', 'Kochi',
       'Visakhapatnam', 'Raipur', 'Ranchi', 'Abu Dhabi', 'Sharjah',
nan],
      dtype=object)
```

MAtches won by more than 100 runs

```
df.head(5)
```

```
   id  season       city        date                          team1  \
0   1    2017  Hyderabad  2017-04-05          Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06               Mumbai Indians
2   3    2017     Rajkot  2017-04-07                Gujarat Lions
3   4    2017     Indore  2017-04-08       Rising Pune Supergiant
4   5    2017  Bangalore  2017-04-08  Royal Challengers Bangalore

                          team2                  toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat

   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
```

```
3                6      GJ Maxwell                      Holkar Cricket
Stadium
4                0       KM Jadhav                       M Chinnaswamy
Stadium

          umpire1        umpire2  umpire3
0     AY Dandekar       NJ Llong      NaN
1  A Nand Kishore         S Ravi      NaN
2     Nitin Menon      CK Nandan      NaN
3    AK Chaudhary  C Shamshuddin      NaN
4             NaN            NaN      NaN
```

```
df.loc[df['win_by_runs']>100,['team1','team2','winner','win_by_runs']]
```

```
                           team1                        team2  \
43                Mumbai Indians             Delhi Daredevils
59         Kolkata Knight Riders  Royal Challengers Bangalore
114             Rajasthan Royals             Delhi Daredevils
295              Kings XI Punjab  Royal Challengers Bangalore
410  Royal Challengers Bangalore                Pune Warriors
556  Royal Challengers Bangalore               Kings XI Punjab
619  Royal Challengers Bangalore                Gujarat Lions

                          winner  win_by_runs
43                Mumbai Indians          146
59         Kolkata Knight Riders          140
114             Rajasthan Royals          105
295              Kings XI Punjab          111
410  Royal Challengers Bangalore          130
556  Royal Challengers Bangalore          138
619  Royal Challengers Bangalore          144
```

Matches Played in Mumbai

```
df[df['city']=="Mumbai"], ['season','team1','team2','winner','city']
```

```
(      id  season    city        date                        team1  \
 6      7    2017  Mumbai  2017-04-09        Kolkata Knight Riders
 9     10    2017  Mumbai  2017-04-12          Sunrisers Hyderabad
 15    16    2017  Mumbai  2017-04-16                Gujarat Lions
 23    24    2017  Mumbai  2017-04-22               Mumbai Indians
 27    28    2017  Mumbai  2017-04-24       Rising Pune Supergiant
 ..   ...     ...     ...         ...                          ...
 572  573    2015  Mumbai  2015-05-19               Mumbai Indians
 576  577    2016  Mumbai  2016-04-09               Mumbai Indians
 584  585    2016  Mumbai  2016-04-16               Mumbai Indians
 589  590    2016  Mumbai  2016-04-20  Royal Challengers Bangalore
 599  600    2016  Mumbai  2016-04-28        Kolkata Knight Riders

                        team2       toss_winner toss_decision  result
```

```
\
 6             Mumbai Indians    Mumbai Indians         field  normal

 9             Mumbai Indians    Mumbai Indians         field  normal

 15            Mumbai Indians    Mumbai Indians         field  normal

 23          Delhi Daredevils  Delhi Daredevils         field  normal

 27            Mumbai Indians    Mumbai Indians         field  normal

 ..                       ...               ...           ...     ...

 572      Chennai Super Kings    Mumbai Indians           bat  normal

 576  Rising Pune Supergiants    Mumbai Indians           bat  normal

 584            Gujarat Lions     Gujarat Lions         field  normal

 589           Mumbai Indians    Mumbai Indians         field  normal

 599           Mumbai Indians    Mumbai Indians         field  normal


      dl_applied                   winner  win_by_runs  win_by_wickets
\
 6             0           Mumbai Indians            0               4

 9             0           Mumbai Indians            0               4

 15            0           Mumbai Indians            0               6

 23            0           Mumbai Indians           14               0

 27            0   Rising Pune Supergiant            3               0

 ..          ...                      ...          ...             ...

 572           0           Mumbai Indians           25               0

 576           0  Rising Pune Supergiants            0               9

 584           0            Gujarat Lions            0               3

 589           0           Mumbai Indians            0               6

 599           0           Mumbai Indians            0               6


     player_of_match             venue          umpire1
umpire2  \
```

```
 6            N Rana  Wankhede Stadium      Nitin Menon       CK
Nandan
 9         JJ Bumrah  Wankhede Stadium      Nitin Menon       CK
Nandan
 15           N Rana  Wankhede Stadium   A Nand Kishore          S
Ravi
 23   MJ McClenaghan  Wankhede Stadium   A Nand Kishore          S
Ravi
 27        BA Stokes  Wankhede Stadium   A Nand Kishore          S
Ravi
 ..              ...                ...               ...              ..
.
 572      KA Pollard  Wankhede Stadium  HDPK Dharmasena  RK
Illingworth
 576       AM Rahane  Wankhede Stadium  HDPK Dharmasena       CK
Nandan
 584        AJ Finch  Wankhede Stadium  HDPK Dharmasena       VK
Sharma
 589       RG Sharma  Wankhede Stadium     AK Chaudhary       CK
Nandan
 599       RG Sharma  Wankhede Stadium      Nitin Menon       RJ
Tucker

      umpire3
 6        NaN
 9        NaN
 15       NaN
 23       NaN
 27       NaN
 ..       ...
 572      NaN
 576      NaN
 584      NaN
 589      NaN
 599      NaN

 [85 rows x 18 columns],
 ['season', 'team1', 'team2', 'winner', 'city'])
```

Matches with no result

```
df[df['winner'].isna()]

      id  season       city        date
team1  \
300  301    2011      Delhi  2011-05-21             Delhi Daredevils

545  546    2015  Bangalore  2015-04-29  Royal Challengers Bangalore
```

```
570  571    2015  Bangalore  2015-05-17             Delhi Daredevils


                           team2                  toss_winner
toss_decision  \
300                Pune Warriors             Delhi Daredevils
bat
545             Rajasthan Royals             Rajasthan Royals
field
570  Royal Challengers Bangalore  Royal Challengers Bangalore
field

        result  dl_applied winner  win_by_runs  win_by_wickets  \
300  no result           0    NaN            0               0
545  no result           0    NaN            0               0
570  no result           0    NaN            0               0

    player_of_match                  venue          umpire1
umpire2  \
300             NaN       Feroz Shah Kotla        SS Hazare     RJ
Tucker
545             NaN  M Chinnaswamy Stadium        JD Cloete     PG
Pathak
570             NaN  M Chinnaswamy Stadium  HDPK Dharmasena  K
Srinivasan

     umpire3
300      NaN
545      NaN
570      NaN
```

Mumbai matches with big win (Win by run more than 100)

```
df[(df['city']=='Mumbai') & (df['win_by_runs']>100)]

      id  season    city        date             team1
team2  \
114  115    2008  Mumbai  2008-05-30  Rajasthan Royals  Delhi
Daredevils

          toss_winner toss_decision  result  dl_applied
winner  \
114  Delhi Daredevils         field  normal           0  Rajasthan
Royals

     win_by_runs  win_by_wickets player_of_match             venue
umpire1  \
114          105               0       SR Watson  Wankhede Stadium  BF
Bowden
```

```
         umpire2  umpire3
114  RE Koertzen      NaN
```

Either abondoned or very close matches win by runs less than 10

```
df[(df['winner'].isnull())| (df['win_by_runs']<10)]
```

```
      id  season       city        date                   team1  \
1      2    2017       Pune  2017-04-06          Mumbai Indians
2      3    2017     Rajkot  2017-04-07           Gujarat Lions
3      4    2017     Indore  2017-04-08  Rising Pune Supergiant
5      6    2017  Hyderabad  2017-04-09           Gujarat Lions
6      7    2017     Mumbai  2017-04-09   Kolkata Knight Riders
..   ...     ...        ...         ...                     ...
629  630    2016     Kanpur  2016-05-21          Mumbai Indians
631  632    2016     Raipur  2016-05-22        Delhi Daredevils
632  633    2016  Bangalore  2016-05-24           Gujarat Lions
634  635    2016      Delhi  2016-05-27           Gujarat Lions
635  636    2016  Bangalore  2016-05-29     Sunrisers Hyderabad

                           team2                  toss_winner
toss_decision  \
1         Rising Pune Supergiant       Rising Pune Supergiant
field
2          Kolkata Knight Riders        Kolkata Knight Riders
field
3                Kings XI Punjab              Kings XI Punjab
field
5            Sunrisers Hyderabad          Sunrisers Hyderabad
field
6                 Mumbai Indians               Mumbai Indians
field
..                           ...                          ...
...
629                Gujarat Lions                Gujarat Lions
field
631  Royal Challengers Bangalore  Royal Challengers Bangalore
field
632  Royal Challengers Bangalore  Royal Challengers Bangalore
field
634          Sunrisers Hyderabad          Sunrisers Hyderabad
field
635  Royal Challengers Bangalore          Sunrisers Hyderabad
bat

     result  dl_applied                       winner  win_by_runs  \
1    normal           0       Rising Pune Supergiant            0
2    normal           0        Kolkata Knight Riders            0
3    normal           0              Kings XI Punjab            0
```

```
5    normal           0          Sunrisers Hyderabad            0
6    normal           0               Mumbai Indians            0
..      ...         ...                          ...          ...
629  normal           0                Gujarat Lions            0
631  normal           0  Royal Challengers Bangalore            0
632  normal           0  Royal Challengers Bangalore            0
634  normal           0          Sunrisers Hyderabad            0
635  normal           0          Sunrisers Hyderabad            8

     win_by_wickets player_of_match  \
1                 7       SPD Smith
2                10         CA Lynn
3                 6      GJ Maxwell
5                 9     Rashid Khan
6                 4          N Rana
..              ...             ...
629               6        SK Raina
631               6         V Kohli
632               4  AB de Villiers
634               4       DA Warner
635               0     BCJ Cutting

                                                 venue          umpire1
\
1             Maharashtra Cricket Association Stadium   A Nand Kishore

2              Saurashtra Cricket Association Stadium      Nitin Menon

3                              Holkar Cricket Stadium     AK Chaudhary

5           Rajiv Gandhi International Stadium, Uppal       A Deshmukh

6                                   Wankhede Stadium      Nitin Menon

..                                               ...              ...

629                                       Green Park     AK Chaudhary

631  Shaheed Veer Narayan Singh International Stadium   A Nand Kishore

632                             M Chinnaswamy Stadium     AK Chaudhary

634                                  Feroz Shah Kotla        M Erasmus

635                             M Chinnaswamy Stadium  HDPK Dharmasena


             umpire2  umpire3
1             S Ravi      NaN
2          CK Nandan      NaN
3      C Shamshuddin      NaN
```

```
5           NJ Llong      NaN
6          CK Nandan      NaN
..               ...      ...
629        CK Nandan      NaN
631     BNJ Oxenford      NaN
632  HDPK Dharmasena      NaN
634        CK Nandan      NaN
635     BNJ Oxenford      NaN

[408 rows x 18 columns]
```

Create a new column margin_type runs>=100 Huge
runs>=50 Big
runs>0 close
else chasing

```
def margin_type(runs):
    if runs>=100:
        return "Huge"
    elif runs>=50:
        return "Big"
    elif runs>0:
        return "Close"
    else:
        return "Chasing"


df['Margin_type']=df['win_by_runs'].apply(margin_type)
df.head()
```

```
   id  season       city        date                        team1  \
0   1    2017  Hyderabad  2017-04-05          Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06               Mumbai Indians
2   3    2017     Rajkot  2017-04-07                Gujarat Lions
3   4    2017     Indore  2017-04-08       Rising Pune Supergiant
4   5    2017  Bangalore  2017-04-08  Royal Challengers Bangalore

                          team2                   toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat
```

```
   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
3               6      GJ Maxwell                      Holkar Cricket
Stadium
4               0       KM Jadhav                       M Chinnaswamy
Stadium

          umpire1        umpire2  umpire3 Margin_type
0     AY Dandekar       NJ Llong      NaN       Close
1  A Nand Kishore         S Ravi      NaN     Chasing
2     Nitin Menon      CK Nandan      NaN     Chasing
3    AK Chaudhary  C Shamshuddin      NaN     Chasing
4             NaN            NaN      NaN       Close
```

How many matches of each margin_type

```
df['Margin_type'].value_counts()
```

```
Margin_type
Chasing    349
Close      239
Big         41
Huge         7
Name: count, dtype: int64
```

Replace city Bangalore with Bengaluru

```
df['city'] = df['city'].replace({'Bangalore':'Bengaluru'})
df.head()
```

```
   id  season       city        date                          team1  \
0   1    2017  Hyderabad  2017-04-05          Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06               Mumbai Indians
2   3    2017     Rajkot  2017-04-07                Gujarat Lions
3   4    2017     Indore  2017-04-08       Rising Pune Supergiant
4   5    2017  Bengaluru  2017-04-08  Royal Challengers Bangalore
```

```
                          team2                  toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat

   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
3               6      GJ Maxwell                      Holkar Cricket
Stadium
4               0       KM Jadhav                       M Chinnaswamy
Stadium

          umpire1        umpire2  umpire3 Margin_type
0     AY Dandekar       NJ Llong      NaN       Close
1  A Nand Kishore         S Ravi      NaN     Chasing
2     Nitin Menon      CK Nandan      NaN     Chasing
3    AK Chaudhary  C Shamshuddin      NaN     Chasing
4             NaN            NaN      NaN       Close
```

Sort matches by season

```
df.sort_values(by = 'season',ascending=False)
```

```
      id  season        city        date                         team1
\
0      1    2017   Hyderabad  2017-04-05          Sunrisers Hyderabad

30    31    2017     Kolkata  2017-04-28             Delhi Daredevils
```

```
32    33    2017        Pune  2017-04-29       Rising Pune Supergiant

33    34    2017      Rajkot  2017-04-29                Gujarat Lions

34    35    2017  Chandigarh  2017-04-30             Delhi Daredevils

..   ...     ...         ...         ...                          ...

92    93    2008      Jaipur  2008-05-11             Delhi Daredevils

91    92    2008   Hyderabad  2008-05-11        Kolkata Knight Riders

90    91    2008     Chennai  2008-05-10          Chennai Super Kings

89    90    2008   Bengaluru  2008-05-28  Royal Challengers Bangalore

101  102    2008   Hyderabad  2008-05-18               Mumbai Indians


                            team2                  toss_winner
toss_decision  \
0    Royal Challengers Bangalore  Royal Challengers Bangalore
field
30         Kolkata Knight Riders        Kolkata Knight Riders
field
32   Royal Challengers Bangalore  Royal Challengers Bangalore
field
33                Mumbai Indians                Gujarat Lions
bat
34               Kings XI Punjab              Kings XI Punjab
field
..                           ...                          ...
...
92              Rajasthan Royals             Rajasthan Royals
field
91               Deccan Chargers        Kolkata Knight Riders
bat
90               Kings XI Punjab              Kings XI Punjab
field
89                Mumbai Indians               Mumbai Indians
field
101              Deccan Chargers              Deccan Chargers
field

     result  dl_applied                  winner  win_by_runs
win_by_wickets  \
0    normal           0     Sunrisers Hyderabad           35
0
30   normal           0   Kolkata Knight Riders            0
```

```
7
32   normal           0  Rising Pune Supergiant           61
0
33      tie           0          Mumbai Indians            0
0
34   normal           0         Kings XI Punjab            0
10
..      ...         ...                     ...          ...
...
92   normal           0        Rajasthan Royals            0
3
91   normal           0   Kolkata Knight Riders           23
0
90   normal           0     Chennai Super Kings           18
0
89   normal           0          Mumbai Indians            0
9
101  normal           0          Mumbai Indians           25
0

    player_of_match                                                venue
\
0      Yuvraj Singh          Rajiv Gandhi International Stadium, Uppal

30        G Gambhir                                        Eden Gardens

32      LH Ferguson            Maharashtra Cricket Association Stadium

33        KH Pandya             Saurashtra Cricket Association Stadium

34   Sandeep Sharma  Punjab Cricket Association IS Bindra Stadium, ...

..              ...                                                ...

92        SR Watson                              Sawai Mansingh Stadium

91       SC Ganguly          Rajiv Gandhi International Stadium, Uppal

90         L Balaji                     MA Chidambaram Stadium, Chepauk

89     CRD Fernando                              M Chinnaswamy Stadium

101        DJ Bravo          Rajiv Gandhi International Stadium, Uppal


                   umpire1         umpire2  umpire3 Margin_type
0              AY Dandekar        NJ Llong      NaN       Close
30                NJ Llong          S Ravi      NaN     Chasing
32   KN Ananthapadmanabhan       M Erasmus      NaN         Big
33            AK Chaudhary     CB Gaffaney      NaN     Chasing
```

```
34                YC Barde       CK Nandan      NaN     Chasing
..                     ...             ...      ...         ...
92                SJ Davis     RE Koertzen      NaN     Chasing
91               IL Howell       AM Saheba      NaN       Close
90          AV Jayaprakash      BG Jerling      NaN       Close
89               BF Bowden  AV Jayaprakash      NaN     Chasing
101            BR Doctrove       DJ Harper      NaN       Close

[636 rows x 19 columns]
```

Total wins per team

```
df['winner'].value_counts()

winner
Mumbai Indians                  92
Chennai Super Kings             79
Kolkata Knight Riders           77
Royal Challengers Bangalore     73
Kings XI Punjab                 70
Rajasthan Royals                63
Delhi Daredevils                62
Sunrisers Hyderabad             42
Deccan Chargers                 29
Gujarat Lions                   13
Pune Warriors                   12
Rising Pune Supergiant          10
Kochi Tuskers Kerala             6
Rising Pune Supergiants          5
Name: count, dtype: int64
```

Which season has biggest average wins

```
df.groupby('season')
['win_by_runs'].mean().sort_values(ascending=False).head(1)

season
2013    16.328947
Name: win_by_runs, dtype: float64



data = {
    "Order_ID": [1, 2, 3, 4, 5, 6, 7, 8],
    "Customer": ["A", "B", "A", "C", "B", "A", "C", "D"],
    "Category": ["Electronics", "Clothing", "Electronics", "Grocery",
"Clothing", "Grocery", "Electronics", "Clothing"],
    "City": ["Bangalore", "Mumbai", "Bangalore", "Delhi", "Mumbai",
```

```
"Bangalore", "Delhi", "Mumbai"],
    "Amount": [20000, 1500, 30000, 800, 2000, 500, 25000, 1800],
    "Quantity": [1, 3, 1, 5, 2, 4, 1, 2]
}

df3 = pd.DataFrame(data)

df3
```

```
   Order_ID Customer     Category       City  Amount  Quantity
0         1        A  Electronics  Bangalore   20000         1
1         2        B     Clothing     Mumbai    1500         3
2         3        A  Electronics  Bangalore   30000         1
3         4        C      Grocery      Delhi     800         5
4         5        B     Clothing     Mumbai    2000         2
5         6        A      Grocery  Bangalore     500         4
6         7        C  Electronics      Delhi   25000         1
7         8        D     Clothing     Mumbai    1800         2
```

Total sales for each category

```
df3.groupby('Category')['Amount'].sum()

Category
Clothing        5300
Electronics    75000
Grocery         1300
Name: Amount, dtype: int64
```

Average order value per city

```
df3.groupby('City')['Amount'].mean()

City
Bangalore    16833.333333
Delhi        12900.000000
Mumbai        1766.666667
Name: Amount, dtype: float64
```

Number of orders per customer

```
df3.groupby('Customer')['Customer'].count()

Customer
A    3
B    2
C    2
D    1
Name: Customer, dtype: int64
```

Maximum order amount per category

```
df3.groupby('Category')['Amount'].max()

Category
Clothing        2000
Electronics    30000
Grocery          800
Name: Amount, dtype: int64
```

Show multiple aggregations together

```
df3.groupby('Category').agg(
    {'Amount':['sum','mean','min','max'],
    'Quantity':['sum','min','mean','max']})


            Amount                                Quantity

                sum          mean    min    max      sum min      mean
max
Category

Clothing      5300   1766.666667   1500   2000        7   2  2.333333
3
Electronics  75000  25000.000000  20000  30000        3   1  1.000000
1
Grocery       1300    650.000000    500    800        9   4  4.500000
5

import pandas as pd

df1 = pd.DataFrame({
    "Name": ["A", "B", "C"],
    "Marks": [80, 90, 85]
})

df2 = pd.DataFrame({
    "Name": ["D", "E"],
    "Marks": [75, 88]
})

print(df1)
print(df2)

  Name  Marks
0    A     80
1    B     90
2    C     85
  Name  Marks
```

```
0    D     75
1    E     88
```

```
pd.concat([df1,df2],axis=0)
```

```
  Name  Marks
0    A     80
1    B     90
2    C     85
0    D     75
1    E     88
```

```
pd.concat([df1,df2],axis=1)
```

```
  Name  Marks Name  Marks
0    A     80    D   75.0
1    B     90    E   88.0
2    C     85  NaN    NaN
```

```
df4 = pd.read_csv(r"C:\Users\uktiw\Downloads\Python\matches -
matches.csv",encoding="Latin")
df4.head(5)
```

```
   id  season       city        date                        team1  \
0   1    2017  Hyderabad  2017-04-05          Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06               Mumbai Indians
2   3    2017     Rajkot  2017-04-07                Gujarat Lions
3   4    2017     Indore  2017-04-08       Rising Pune Supergiant
4   5    2017  Bangalore  2017-04-08  Royal Challengers Bangalore

                          team2                  toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat

   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
```

```
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
3               6      GJ Maxwell                      Holkar Cricket
Stadium
4               0       KM Jadhav                       M Chinnaswamy
Stadium

          umpire1        umpire2  umpire3
0     AY Dandekar       NJ Llong      NaN
1  A Nand Kishore         S Ravi      NaN
2     Nitin Menon      CK Nandan      NaN
3    AK Chaudhary  C Shamshuddin      NaN
4             NaN            NaN      NaN
```

```
pip install openpyxl
```

```
Requirement already satisfied: openpyxl in c:\users\uktiw\anaconda3\
lib\site-packages (3.1.5)
Requirement already satisfied: et-xmlfile in c:\users\uktiw\anaconda3\
lib\site-packages (from openpyxl) (2.0.0)
Note: you may need to restart the kernel to use updated packages.
```

```
df5 = pd.read_excel(r"C:\Users\uktiw\Downloads\Python\Country-
Code.xlsx")
df5.head()
```

```
   Country Code    Country
0             1      India
1            14  Australia
2            30     Brazil
3            37     Canada
4            94  Indonesia
```

```
df6=pd.merge(df4,df5, on= "Country Code", how= "left")
df6
```

```
      Restaurant ID           Restaurant Name  Country Code
City  \
0           6317637          Le Petit Souffle           162
Makati City
1           6304287          Izakaya Kikufuji           162
Makati City
2           6300002    Heat - Edsa Shangri-La           162
Mandaluyong City
3           6318506                      Ooma           162
Mandaluyong City
```

```
4           6314302                Sambo Kojin           162
Mandaluyong City
...             ...                        ...           ...
...
9546        5915730               NamlÛ± Gurme           208
ÛÁstanbul
9547        5908749              Ceviz AÛôacÛ±            208
ÛÁstanbul
9548        5915807                     Huqqa            208
ÛÁstanbul
9549        5916112               A)ô)ôk Kahve           208
ÛÁstanbul
9550        5927402  Walter's Coffee Roastery            208
ÛÁstanbul

                                                  Address  \
0     Third Floor, Century City Mall, Kalayaan Avenu...
1     Little Tokyo, 2277 Chino Roces Avenue, Legaspi...
2     Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...
3     Third Floor, Mega Fashion Hall, SM Megamall, O...
4     Third Floor, Mega Atrium, SM Megamall, Ortigas...
...                                                 ...
9546  Kemanke)ô Karamustafa Pa)ôa Mahallesi, RÛ±htÛ±...
9547  Ko)ôuyolu Mahallesi, Muhittin íìstí_ndaÛô Cadd...
9548  Kuruí_e)ôme Mahallesi, Muallim Naci Caddesi, N...
9549  Kuruí_e)ôme Mahallesi, Muallim Naci Caddesi, N...
9550  CafeaÛôa Mahallesi, BademaltÛ± Sokak, No 21/B,...

                                          Locality  \
0      Century City Mall, Poblacion, Makati City
1     Little Tokyo, Legaspi Village, Makati City
2     Edsa Shangri-La, Ortigas, Mandaluyong City
3         SM Megamall, Ortigas, Mandaluyong City
4         SM Megamall, Ortigas, Mandaluyong City
...                                          ...
9546                                   Karakí_y
9547                                  Ko)ôuyolu
9548                                Kuruí_e)ôme
9549                                Kuruí_e)ôme
9550                                       Moda

                                        Locality Verbose   Longitude  \
0     Century City Mall, Poblacion, Makati City, Mak...  121.027535
1     Little Tokyo, Legaspi Village, Makati City, Ma...  121.014101
2     Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...  121.056831
3     SM Megamall, Ortigas, Mandaluyong City, Mandal...  121.056475
4     SM Megamall, Ortigas, Mandaluyong City, Mandal...  121.057508
...                                                 ...         ...
9546                                Karakí_y, ÛÁstanbul   28.977392
9547                               Ko)ôuyolu, ÛÁstanbul   29.041297
```

```
9548                              Kuruí_e)ôme, ÛÁstanbul   29.034640
9549                              Kuruí_e)ôme, ÛÁstanbul   29.036019
9550                                     Moda, ÛÁstanbul   29.026016

       Latitude                           Cuisines  ...  Has Table
booking  \
0     14.565443        French, Japanese, Desserts  ...
Yes
1     14.553708                           Japanese  ...
Yes
2     14.581404  Seafood, Asian, Filipino, Indian  ...
Yes
3     14.585318                    Japanese, Sushi  ...
No
4     14.584450                   Japanese, Korean  ...
Yes
...         ...                                ...  ...
...
9546  41.022793                           Turkish  ...
No
9547  41.009847   World Cuisine, Patisserie, Cafe  ...
No
9548  41.055817            Italian, World Cuisine  ...
No
9549  41.057979                    Restaurant Cafe  ...
No
9550  40.984776                               Cafe  ...
No

     Has Online delivery Is delivering now Switch to order menu Price
range  \
0                     No                 No                    No
3
1                     No                 No                    No
3
2                     No                 No                    No
4
3                     No                 No                    No
4
4                     No                 No                    No
4
...                  ...                ...                   ...
...
9546                  No                 No                    No
3
9547                  No                 No                    No
3
9548                  No                 No                    No
4
```

```
9549                   No                 No                    No
4
9550                   No                 No                    No
2

     Aggregate rating  Rating color  Rating text Votes      Country
0                 4.8    Dark Green    Excellent   314  Phillipines
1                 4.5    Dark Green    Excellent   591  Phillipines
2                 4.4         Green    Very Good   270  Phillipines
3                 4.9    Dark Green    Excellent   365  Phillipines
4                 4.8    Dark Green    Excellent   229  Phillipines
...                ...           ...          ...   ...          ...
9546              4.1         Green    Very Good   788       Turkey
9547              4.2         Green    Very Good  1034       Turkey
9548              3.7        Yellow         Good   661       Turkey
9549              4.0         Green    Very Good   901       Turkey
9550              4.0         Green    Very Good   591       Turkey

[9551 rows x 22 columns]
```